

The Effect of Minimum Wages on Actual Wages in Formal and Informal Sectors in Costa Rica

T.H. GINDLING

University of Maryland at Baltimore County, USA

and

KATHERINE TERRELL *

University of Michigan, Ann Arbor, USA

Summary. — As in many economies, wage gaps between formal and informal sectors have been persistent in Costa Rica. Using 12 years of microdata, we test whether increases in minimum wages raise actual wages; we find they not only raise wages in large urban and rural enterprises (traditionally referred to as formal), but they also raise wages of workers covered by minimum wage legislation in what are traditionally regarded as informal sectors where the legislation is often considered not to be enforced: small urban and small rural enterprises. Furthermore, minimum wages raise wages in these informal sectors more than in larger firms and hence work to reduce wage differentials between the larger and smaller firms. However, minimum wages have no significant impact on wages in the informal sector not covered by minimum wage legislation, the self-employed.

© 2005 Elsevier Ltd. All rights reserved.

Key words — informal sector, minimum wages, wages, Costa Rica, Latin America

1. INTRODUCTION

The degree to which legal minimum wages in developing countries have an impact on actual wages is of substantial interest to analysts and policy makers in both developing and developed countries. First, as Freeman (1993) and McLeod and Lustig (1997) argue, legal minimum wages have the potential to reduce the level of poverty within developing countries. However, if minimum wages are enforced only in the relatively high-wage formal sectors, they are unlikely to help workers in the parts of the economy where most of the poor are found (the rural and urban informal sectors). Second, compared to the developed economies, minimum wages in developing economies tend to be set at higher levels relative to the average wages. Some observers are therefore worried that high minimum wages may have a negative effect on employment in the formal sector, pushing workers into and thus lowering wages in lower-paid informal sectors (World Bank,

* We would like to thank two anonymous referees for comments that greatly improved the paper and Rebecca Blank, Charlie Brown, John DiNardo, Francisco Ferreira, Gustavo Gonzaga, Daniel Hamermesh, John Kennan, Janos Kollo, Sara Lemos, Martin Ravallion, Brad Humphreys, Pablo Sauma, Jan Svejnar, Juan Diego Trejos, and Juan Rafael Vargas for helpful suggestions on earlier versions of this paper. Staff members at the Costa Rican Central Bank and Ministry of Labor and Social Security were very helpful in providing us with data and important information on minimum wages. We are especially grateful to José Pablo Carvajal, Orlando Garcia, and Yabera Alvarado of the Ministry of Labor. We benefited from comments of seminar participants at the Costa Rican Central Bank (June 5, 2002), the Network for Inequality and Poverty Meetings in Madrid (November 10, 2002), the World Bank Applied Microeconomics seminar (December 6, 2002), and the University of Maryland Baltimore County (October 15, 2003). Justine Wagner provided invaluable research assistance. Katherine Terrell would like to acknowledge the support from the NSF (grant SES-0111783). Final revision accepted: April 15, 2005.

1990, p. 63). Third, from the point of view of labor unions in the developed economies, the low level of actual and minimum wages in developing countries constitutes an unfair advantage in international trade. As a result, unions have pushed for including higher minimum wage targets for developing countries in bilateral and multilateral trade agreements. The degree to which all of these points of view are good representations of reality depends, in part, on the degree to which legal minimum wages have an impact on the actual wages of workers, and in which sectors. In this paper, we address these issues.

We estimate the effect of increases in minimum wages on the wages of workers in formal and informal sectors, using two definitions of formality. The literature has broadly distinguished these sectors on the basis of differential enforcement or differential coverage of regulations. It is generally assumed that minimum wages are enforced in larger firms and among unionized workers sector, but that they are enforced weakly or not at all in the small rural and urban sectors. As *Ranis and Stewart (1999, p. 260)* put it, "In most less developed countries, governments with limited administrative resources tend to focus on large-scale operations when trying to collect taxes or enforce minimum wage legislation." Hence, for our first definition, we will use firm size and urban/rural dichotomy to distinguish the formal from the informal sector. Our second distinction of the two sectors is based on legal coverage. In Costa Rica, minimum wage legislation covers all employees, irrespective of firm size or place of work. Hence, the informal or uncovered sector is the self-employed workers.

Indeed, differential enforcement of or coverage by minimum wage legislation has been cited as a key reason for the existence of dualistic labor markets in developing economies. *Ranis and Fei (1961, p. 536)* note that the persistence of the urban/rural wage gap assumed by *Lewis (1954)* must be "sustained by institutional or nonmarket forces since under competitive assumptions real wages [in the formal sector] would fall to zero, at equality with the MPP [the marginal physical product] in agriculture." *Harris and Todaro (1970, p. 126)* also ascribe the urban/rural wage gap to "the existence of a politically determined minimum urban wage at levels substantially higher than agricultural earnings." *Fields (1975)* extends the concept of economic dualism to the urban sector, arguing that differential enforcement of legal

minimum wages can lead to an institutionally determined high-wage in the urban formal sector and a lower competitive wage in the urban informal sector.

By examining whether minimum wages are enforced in what is usually considered to be the informal sector, we provide evidence that is important for analyzing the issue of formal-informal wage gaps among covered sector workers. If legal minimum wages are enforced only in the urban large-scale sector, for instance, we would expect that higher minimum wages directly result in higher wages in the urban formal sector but do not directly raise the wages of workers in the rural sector or in small firms in the short term (even though minimum wages may be legally applicable to these workers). On the other hand, if minimum wages are enforced in all sectors where they are legally applicable, we would expect to find that higher legal minimum wages result in the short term in higher wages for all types of employees covered by minimum wage law, whether they work in large firms, small firms, or in rural areas. Furthermore, in this case, we would expect higher minimum wages to have a larger impact on the wages of workers in those sectors, where more workers earn wages that are near the minimum wage. In particular, we would expect higher minimum wages in this case to have a larger effect on the wages of workers in the rural sector and in small firms than in the urban formal sector.

Whereas predictions about the impact on the formal-informal wage gaps of better enforcement of minimum wages in the legally covered sector are fairly straightforward, the impact of minimum wages on the wage gaps in the covered and uncovered sectors is more complex. An increase in the minimum wage will not have a direct effect on the sectors, where it is not covered or enforced. Nevertheless, over time, the change in the minimum wage may be expected to affect the uncovered (and unenforced) sector. In the classical dual sector model, workers in the formal sector (covered by a binding minimum wage) who lose jobs as a result of increases in the minimum wage will seek employment in the informal (uncovered or imperfectly covered) sector, thus increasing the labor supply and lowering wages in the informal sector. This implies that an increase in the minimum wage in the covered/formal sector increases the wage gap between the formal and informal sectors. One problem with this model is the assumption of no open unem-

ployment. Mincer (1976) has shown that when unemployment is allowed in this type of a model, a rise in minimum wages can lead to a rise in both the covered and uncovered sector wages if there is a sufficient increase in unemployment.¹ The Todaro (1969) and Saint-Paul (1994) models also predict that increases in minimum wages in the urban area can indirectly raise rural wages. The mechanism in the Todaro model is a backward shift of labor supply in the rural area as workers leave for the urban sector to seek higher-wage formal-sector jobs. In the model of Saint-Paul (1994), wages in the rural area rise as a result of an increase in the demand for labor brought about by capital flows from urban to rural areas in response to higher wages (triggered by higher minimum wages) in the urban area. More recently, McIntyre (2004) has a model, which reflects Brazilian institutions and has workers deciding between formal sector employment with benefits and lower wages and informal sector employment with no benefits and higher wages. In this model, an increase in minimum wages can increase wages in both sectors. Hence, these and other models (e.g., Fiszbein, 1992) demonstrate that raising the minimum wage in the covered sector may in the longer run have positive or negative effects on wages in the uncovered sector, with the outcome depending on a number of factors that may shift the supply and demand curves for labor and determine their elasticities. As this discussion indicates, one would not expect an increase in the minimum wage to affect directly wages in the sector that is not covered by minimum wage law in Costa Rica, namely the self-employed but the long-run effect on their wages, and on the gap between their wages and the covered sector, is ambiguous.

To estimate the effect of minimum wages on wages, we use data from Costa Rica, a developing economy with a complex legal minimum wage structure, which applies to all employees, irrespective of whether they work in a rural or urban area, in large or small firms, part time or full time. The only workers not covered by minimum wage protection are the self-employed. An aspect of the Costa Rica minimum wage structure, which makes it valuable for research, is that the changes in the structure of minimum wages over the 1988–99 period resulted in variation in the minimum wages over time and within occupations that were exogenous to changes in the labor market. Because we use these frequent (at least once a year)

exogenous variations to estimate the impact of minimum wages on wages, our results do not suffer from the potential endogeneity bias plaguing most previous studies.

We use the entire set of multiple legal minimum wages each year to estimate the impact of minimum wages on wages. Multiple minimum wages are not uncommon in Latin America. In Mexico, for example, minimum wages are set separately for 3 regions and 88 occupations, while in Honduras there are over 24 minimum wages (for two firm sizes in each of 12 industries, plus separate minimum wages for the export sectors). It is therefore surprising that this and two other papers we have written (Gindling & Terrell, 2005a, 2005b) are the first studies that we know of that takes advantage of the full complexity of the minimum wage structure to estimate the impact of minimum wages on actual wages and employment in Latin America. The other empirical analyses make the simplifying assumption that only one minimum wage (generally the lowest) applies to all workers.

Using annual data from the Costa Rican Household Surveys for the 1988–99 period, we find that (a) the gaps in the wages of formal and informal sector workers are persistent, (b) increases in minimum wages raise directly the wages of workers in the large urban enterprises as well as workers covered by minimum wage legislation in what are traditionally regarded as informal sectors and where the legislation is often assumed not to be enforced, and (c) increases in minimum wages do not have a direct, statistically significant effect on the wages of workers not covered by the minimum wage legislation.

2. LITERATURE REVIEW

The literature on the impact of legal minimum wages on the wages of workers in developing countries is sparse and the evidence is almost entirely on the effect for workers in the formal sector. However, recently a number of studies have appeared which use data for Latin American countries and explore the impact on informal sector wages. Collectively they indicate that increases in minimum wages have a positive effect on the wages of workers in the formal sector but the impact on the informal sector is ambiguous. We briefly review the findings using various methodologies in the Latin American literature.

Many studies examine the issue of the impact of minimum wages by looking for spikes in the distribution of wages at or near the legal minimum wage. Here the evidence for Latin American countries is mixed. [Bell \(1997\)](#) finds a spike in the distribution of wages that corresponds to the minimum wage in Colombia but not in Mexico. [Maloney and Núñez \(2004, chap. 1\)](#) find spikes at the minimum wage for workers in the formal sector in Brazil, Chile, Colombia, Brazil, and Honduras, but not in Argentina, Mexico, or Uruguay. On the other hand, spikes have been found near the minimum wage in Brazil at various times ([Fajnzylber, 2001](#); [Lemos, 2002](#)), Puerto Rico at the time the minimum wage was being raised to the US level ([Castillo-Freeman & Freeman, 1992](#)), and in Trinidad and Tobago when the minimum wage was introduced ([Strobl & Walsh, 2003](#)). The impact of a minimum wage on the distribution of wages depends on both its level and enforcement and these vary across countries and over time.

The evidence from traditional econometric studies generally indicates that increases in minimum wages increase wages in the formal (large-scale) sector when the minimum wages are binding. [Bell \(1997\)](#) uses time-series data on medium- and large-scale manufacturing firms to regress the log of average wages in a firm on the log of minimum wages, and other covariates, for Mexico and Colombia. She finds that minimum wages did not have a significant impact on wages in Mexico, where minimum wages were relatively low and falling throughout the period, but they did have a significant impact (with an elasticity of 0.37) in Colombia, where they were relatively high and rising. [Maloney and Núñez \(2004\)](#) and [Fajnzylber \(2001\)](#) find large significant effects of minimum wages on wages of formal sector employees initially earning within 10% of the minimum wage: [Maloney and Núñez \(2004\)](#) report an elasticity of 0.6 for Colombia, while [Fajnzylber \(2001\)](#) reports an elasticity of 1.08 for Brazil.² More recently, [Strobl and Walsh \(2003\)](#) use panel data to estimate whether the minimum wage affects the probability that a worker initially earning below the minimum wage would earn more than the minimum wage after the introduction of the national minimum wage in Trinidad and Tobago in 1998. They find that this probability increased by about 20% and more important for this study, this effect is bigger for workers in large firms (10 or more employees) compared to small firms.

There are even fewer studies using Latin American data that examine the impact of minimum wages on the wages of workers in the informal sector. [Maloney and Núñez \(2004\)](#) define the informal sector in Colombia as the self-employed; [Fajnzylber \(2001\)](#) presents estimates for two definitions of the informal sector in Brazil: self-employed and unregistered employees; and [Lemos \(2002\)](#) uses the unregistered employee definition in Brazil. [Maloney and Núñez \(2004\)](#) find that minimum wages do not have a significant effect on the wages of self-employed workers near the minimum wage (but are very large and significant for workers initially earning less than 70% of the minimum wage) while [Fajnzylber \(2001\)](#) finds that the impact on the wages of informal sector workers is not only significant, but larger than the impact on formal sector workers: he estimates an elasticity of 1.03 for unregistered paid employees and 1.32 for self-employed workers. [Lemos \(2002\)](#) using the same data, but a different methodology also finds a larger elasticity on low-wage workers in the informal sector than in the formal sector in Brazil.

[Fajnzylber \(2001\)](#) argues that even though legal minimum wages are not applicable to the informal sector, the positive wage effect represents a “lighthouse effect” whereby informal sector workers set their wages by referring to the legal minimum wage. This explanation is possible in Brazil where “unregistered workers” can select between registered and unregistered status. As [Maloney and Núñez \(2004\)](#) and [McIntyre \(2004\)](#) note, the Brazilian workers may be trading off a package of lower wages plus benefits in registered employment for higher wages and no benefits in unregistered (informal) employment. However, it is not clear why this should be the case with self-employed workers, unless they work as independent contractors for these firms.

An alternative explanation for the large wage effects in the informal (uncovered) sector, which can also apply to the formal sector, is that changes in legal minimum wages and changes in wages over time may both be driven by some other unmeasured phenomenon, for example, inflation. The endogeneity problem is indeed serious in this literature and the current generation of studies has been paying particular attention to this issue. [Strobl and Walsh \(2003, p. 428\)](#) for instance note that the introduction of a national minimum wage in Trinidad and Tobago was largely unanticipated, and thus “can be argued to be at least weakly

exogenous.” We note below that the administrative changes in the structure of minimum wages in Costa Rica result in variation in the minimum wages, over time and within occupations, that were exogenous to changes in the labor market. Therefore, the results presented in our present paper suffer less from the potential endogeneity bias that may be affecting the estimates of other papers in this literature.

3. DATA AND METHODOLOGY

For our analysis, we combine information on legal minimum wages from the decrees, published by the Ministry of Labor, with data on individuals from the annual *Household Surveys for Multiple Purposes*, carried out by the Costa Rican Institute of Statistics and Census. The household surveys have been conducted in July of every year since 1976 on approximately 1% of the population. They include questions on earnings; employment; hours worked; job characteristics such as industry, occupation, sector, and firm size; as well as education, age; and many other personal characteristics of the individual in the household. We create a cross-section/time-series data set for all individuals who worked in the private sector.³ In this paper, we use data from 1988 onward because in these years, the occupation categories in the household surveys are sufficiently detailed to enable us to match them with the detailed occupation/skill/industry categories in the minimum wage decrees. Overall, we use data on an average of approximately 10,000 workers in each year for 12 years (1988–99).

Figure 1 presents macroeconomic statistics on the mean real hourly minimum wage from the Costa Rican Household Surveys (weighted by the number of workers in each minimum

wage category), the annual rates of GDP growth, and inflation (measured by the CPI) for each year from 1988 to 1999. We note that the average real minimum wage moves somewhat with the business cycle and inflation rates for most of the period. The exception are 1996 and 1997, when the minimum wage grew at a more rapid rate, perhaps to make up for the fall in the real value of minimum wages in earlier years (e.g., 1990, 1993, and 1994). Average annual GDP growth is robust for most of the period, but weak in 1991 (2.3%) and 1996 (0.3%) when inflation, which is at double digits for most of the period, peaked at 29% (in 1991) and 23% (in 1996).

In this paper, we examine the effect of changes in minimum wages on the wages of workers in various sectors, which are often identified as the formal and informal sectors. We use two definitions of formality: Our first definition is the one most commonly used by the International Labor Office (ILO) and the UN Regional Employment Program for Latin America and the Caribbean (widely known by its Spanish acronym, PREALC). According to the ILO–PREALC definition, the formal sector includes all employees in large urban or rural firms/farms plus all workers with a university education or who are classified as professional or technical employees. The informal sector is defined as those who work in firms with five or fewer employees (and do not have a university education or are classified as professional or technical employees) or are self-employed.⁴

A second way of identifying the informal sector is to define it as the sector where workers are not covered by labor regulations. For example, Maloney and Núñez (2004) define workers in the informal sector in Mexico as those that do not pay social security and Fajnzylber (2001) defines them in Brazil as the unregistered

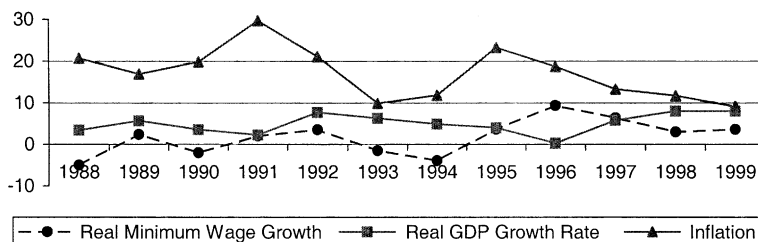


Figure 1. Macroeconomic statistics. Notes: GDP growth and inflation (from CPI) from Banco Central de Costa Rica; Average Real Minimum Wage Growth calculated from the Costa Rican Household Survey data using sample weights.

employees who do not receive social security and other benefits. Within this framework, one could also identify informal sector workers as those that are not protected by minimum wage legislation (uncovered workers). The actual definition implemented in a study is obviously constrained by data limitations. We do not have information on registration or social security payments in our data and but we are able to distinguish workers covered by minimum wage legislation (all employees in the private sector) from those not covered (the self-employed).

Within each of these definitions, we also distinguish between work in the urban and rural areas since it is generally believed that enforcement of labor legislation is weaker in rural areas than in urban. In summary, these two definitions of informality yield six sectors, which we use in our analysis: the urban and rural formal (large enterprise) sectors, two informal sectors where minimum wages are leg-

ally applicable to workers (the urban and rural small firm sectors), and two informal sectors where workers are not covered by minimum wages (urban and rural self-employed).

We begin with an examination of the wage gaps among these groups of workers. Table 1 presents pairwise differences in three-year average log real wages across these six sectors in 1988–90 and 1997–99. The evidence is stark: wage gaps are significant and persistent for all pairs in both of these time periods. Not surprisingly, the largest gap is between the urban formal (large-scale) and rural informal (small-scale) sectors (about 0.77 log points) followed by the gap between the urban large and urban small-scale sectors (about 0.59 log points). We also find that the mean wage of the urban self-employed is significantly higher than the mean wage of all sectors except for the urban large-scale sector.

The relatively high mean earnings of the urban self-employed should not be surprising

Table 1. *Real three-year average wage gaps between sectors (Sector 1 minus Sector 2)*

Sector 1	Sector 2					
	Covered sector			Uncovered sector		
	Urban large	Urban small	Rural large	Rural small	Urban self-employed	Rural self-employed
<i>1988–90</i>						
Urban large	0	0.59***	0.36***	0.78***	0.12	0.49***
Urban small		0	–0.23***	0.20***	–0.47***	–0.11***
Rural large			0	0.43***	–0.24***	0.13***
Rural small				0	–0.68***	–0.31***
Urban self-employed					0	0.37***
Rural self-employed						0
<i>1997–99</i>						
Urban large	0	0.60***	0.36***	0.76***	0.19	0.46***
Urban small		0	–0.25***	0.15***	–0.41***	–0.14***
Rural large			0	0.40***	–0.16***	0.11***
Rural small				0	–0.56***	–0.29***
Urban self-employed					0	0.27***
Rural self-employed						0
<i>Change in gap from 1988–90 to 1997–99</i>						
Urban large	0	0.01	0.00	–0.04**	0.07*	–0.02
Urban small		0	–0.01	–0.05*	–0.06**	0.03
Rural large			0	–0.03**	–0.08***	–0.02
Rural small				0	–0.11***	–0.02
Urban self-employed					0	–0.09***
Rural self-employed						0

Notes: Calculated from log wages using sample weighted data from the *Costa Rican Household Survey*.

* Significantly different from 0 at 10%.

** Significantly different from 0 at 5%.

*** Significantly different from 0 at 1%.

since substantial evidence in the literature, as well as variance around these mean wages, indicates that the self-employed is a very heterogeneous group including successful owners of small businesses as well as destitute workers eking out earnings on the street.⁵ In fact, over the 1988–99 period, approximately one-fifth of the self-employed in Costa Rica are small-scale entrepreneurs (employers with one to three employees). It seems that in Costa Rica, at least, the self-employed are doing better on average than many other workers. Hence, this may be a “sector of choice” for many, as Maloney (2004) argues using data for other Latin American countries, and not a “sector of last resort” as was traditionally assumed in many of the early dual economy models.

The data underlying these log wage gaps are presented in Table 2, along with three-year average minimum wages and the distribution of workers in the six formal/informal sectors defined above. Just as the wage gaps are persistent, the distribution of workers in these six categories has not changed appreciably over time, signaling potentially little mobility across these categories. If minimum wages play a role in the gaps among covered sector workers, they would need to be higher in those sectors where wages are higher. We can see this is the case in Table 2: the average real minimum wage is highest in the urban large-scale sector and lowest in the urban small and rural small-scale. However, relative to wages in each of the sectors, the minimum wages for the rural small enterprise sector are the highest (about 95–99% of the average wage) and the minimum

wage for the urban large-scale sector is the lowest (56–64% of the average wage). This suggests that if legal minimum wages disproportionately affect the wages of low-wage workers, we would expect to find a bigger impact of minimum wages on the wages of workers in rural areas and small enterprises than in the urban formal sector. We note that the average real minimum wage that is applicable to the occupations in which the self-employed are found is relatively low compared to their salaries.

Whereas enforcement of minimum wages in the low-wage sector can help close the gap among covered sector workers over time, it would be more effective if in addition the minimum wage were raised more rapidly for the low-wage sectors relative to the high-wage sectors. However, this may be too stringent of an expectation since one would expect the high-wage workers would have more of a voice and be able to increase their minimum wages more rapidly. We find that indeed this is not the case in Costa Rica since the minimum wage is rising most rapidly in the urban large-scale sector.

Our analysis of the impact of minimum wages on the wages of workers in the formal and informal sectors is based on two methods: (a) a comparison of the distribution of minimum wages and actual wages, looking for spikes in actual wages around the minimum wages and (b) an econometric analysis of wages and minimum wages. We begin by comparing kernel density functions of the wages of workers with kernel density functions of the distribution of minimum wages.⁶ If minimum wages

Table 2. *Distribution of workers, real hourly wages and real hourly minimum wages (1999 colones), by sector, three-year averages*

Sector	Percent or workers in each sector		Real wage		Real minimum wages		Average minimum wage/ average wage	
	1988–90	1997–99	1988–90	1997–99	1988–90	1997–99	1988–90	1997–99
<i>Covered sectors</i>								
Urban large	16.35	15.37	530 (9.44)	631 (8.24)	295 (1.49)	401 (1.95)	0.56	0.64
Urban small	7.33	7.10	281 (8.34)	323 (6.84)	216 (1.59)	260 (1.38)	0.77	0.80
Rural large	29.08	27.43	332 (5.98)	408 (4.83)	255 (0.56)	324 (1.00)	0.77	0.79
Rural small	18.11	18.66	219 (2.79)	274 (3.03)	217 (0.80)	261 (0.63)	0.99	0.95
<i>Uncovered sectors</i>								
Urban self-employed	10.24	10.64	513 (14.73)	609 (24.58)	257 (0.51)	295 (0.79)	0.50	0.48
Rural self-employed	18.89	20.86	373 (7.78)	477 (11.29)	245 (0.30)	235 (0.41)	0.66	0.49

Notes: Calculated from the *Costa Rican Household Survey* using sample weighted data. Standard errors in parentheses.

have an effect, we should see spikes in the wage distribution around the minimum wages. However, evidence of spikes cannot be interpreted as being caused by minimum wages since there are other possible reasons as to why minimum wages and actual wage spikes may coincide. For example, both minimum wages and the wages reported by survey recipients may cluster at round numbers, wages may be standardized for sizeable trade or occupation groups, or actual wages and minimum wages may cluster around discrete levels of human capital.

To provide additional evidence of the impact of minimum wages on actual wages in each sector, we estimate the extent to which changes in the minimum wage affect wages, holding constant other factors that might affect wages, by using individual-level pooled cross-section/time-series data for 1988–99. Specifically, we estimate separately for each sector an equation of the form:

$$\ln W_{it} = \alpha_0 + \eta \ln MW_{it} + X'_{it}\beta + \sum_{j=1}^J \lambda_j OCC_{itj} + \sum_{t=1}^T \gamma_t YR_t + \mu_{it}, \quad (1)$$

where the dependent variable, $\ln W_{it}$, is the log of the real hourly wage (in 1999 colones) of individual i in year t (1988–99). The explanatory variables include the log of the real minimum wage (in 1999 colones) that applies to that worker's occupation category in each year, $\ln MW_{it}$. The coefficient η is an estimate of the impact on average actual wages of changes in the legal minimum wage. Other explanatory variables include the vector X_{it} , of individual specific human capital variables (years of education, a quartic in experience, gender, and full interactions among these variables). We also include dummy variables for each minimum wage (occupation/skill) categories, OCC_{itj} ($j = 1, \dots, 350$), in order to control for category-specific fixed effects and for the endogenous cross-sectional correlation of wages and minimum wages across minimum wage categories.⁷ Finally, to control for endogenous changes in yearly average minimum wages (as well as other year-specific factors such as aggregate supply and aggregate demand changes, the timing of minimum wage changes,⁸ or design changes in the household surveys) we include a dummy variable for each year, YR_t ($t = 1989-99$). After including the dummy variables for occupation and years, our resulting estimates of the impact of legal minimum wages

on wages are based on deviations of each minimum wage from the average minimum wage change within occupation categories over time. We next argue that these changes can reasonably be thought of as exogenous.

A major problem plaguing the empirical minimum wage literature is one of endogeneity bias that arises if minimum wages are set according to changes in demand and supply conditions, which also affect wages. This occurs, for example, when the level of minimum wages is adjusted by the amount of inflation in the previous period, a practice followed in Costa Rica and common in many countries. However, we argue that a special feature of Costa Rica's minimum wage policy over this period implies that we do not have a simultaneity problem in our estimations.⁹ During the period under study, the government of Costa Rica implemented a policy of gradually reducing the number of minimum wages from over 500 categories (set by the industry and occupation of the worker) in 1988 to 19 categories (set by skill only) in 1997. The process of simplification made changes in the relative minimum wage within occupational categories exogenous over this period because these changes occurred solely in order to harmonize wages within an occupation category across industries. The Ministry identified broadly defined occupation categories to be harmonized across industries and proceeded, gradually over a period of several years, to increase the lower(est) minimum wage by a greater amount than the higher(est) minimum wage within each broadly defined occupational category. Over a period of several years, one minimum wage emerged for each broadly defined occupation, irrespective of industry. Effectively, this increased the amount of exogenous year-to-year variation in minimum wages because the minimum wages for different occupation categories in each year were increased by different amounts.

Another source of exogenous variation in minimum wages was the administrative decision to increase the number of minimum wages for workers with higher education. In 1988, a minimum wage was set for all workers with a five-year university degree (*licenciado*), irrespective of occupation. In 1993, a new minimum wage was set for individuals with two to three years of university education (*diplomados*) and for graduates of five-year technical high schools (*técnicos*). In 1997, another new minimum wage was added for workers with a four-year university degree. By 1997, there were 19 minimum

wages, one each for unskilled workers, semi-skilled workers, skilled workers, specialized workers (supervisors), and domestic servants, and several for professionals (without any industrial dimension).

Hence, whereas an underlying criterion for setting the average minimum wage was the rate of inflation (measured by the consumer price index) in the preceding six months, the minimum wage for each occupation/skill category was increased at different rates around this average and these rates did not depend on demand conditions for that specific occupation. Therefore, we argue that after controlling for the average change in the minimum wage by year (which we do with a set of year-specific dummy variables) and for cross-sectional variation across minimum wage categories (which we do with a set of minimum wage category-specific dummy variables), any remaining variation in legal minimum wages is exogenous to demand and supply conditions in the labor market, and therefore exogenous to actual wage changes.

As an additional test of the exogeneity of the minimum wage variable, we re-estimated our equations using lagged minimum wages as an instrumental variable. In the reduced form equations, lagged minimum wages were positively correlated with actual minimum wages at the 1% significance level for all sectors.¹⁰ To test for the endogeneity of the minimum wage variable, we then included the residual from the reduced form equation as an additional explanatory variable in Eqn. (1). The results of these regressions provide no evidence of endogeneity of the minimum wage variable in any of the covered sectors. For all sectors, except for the urban self-employed sector, the coefficient on the residual was not significantly different from zero at the 10% significance level.¹¹ We are hence reasonably confident that our results do not suffer from the endogeneity/simultaneity bias that may exist in the studies that compare changes in a single minimum wage to changes in actual wages.

4. FINDINGS

(a) *Distribution of wages and minimum wages*

We begin by looking for spikes in the distribution of wages that are at the minimum wages. Figures 2–4 overlay the distribution of

wages on the distribution of minimum wages in 1988 and 1999 for the urban covered sector (Figure 2), the rural covered sector (Figure 3) and the uncovered rural and urban sector (Figure 4). Spikes in the distribution of minimum wages represent legal minimum wages that apply to a large proportion of workers. For example, in 1988, the first spike is at the minimum wage for domestic servants, who represent approximately 7% of all workers and to whom applies a legal minimum wage of 123 colones (1999 prices) or \$0.43 (1999 US dollars) per hour. The second spike, at 231 colones (or \$0.81 per hour), is at the minimum wage for unskilled workers which represents over 20% of all workers in 1988. At the very right of the distribution of minimum wages is a spike at the minimum wage for university graduates (*licenciados*) who represent approximately 2% of all workers and to whom applies a legal minimum wage at 578 colones or \$2.00 per hour.

Several important points are immediately clear from observing these graphs. First, legal minimum wages in Costa Rica are applicable to workers across the wage distribution. Minimum wages are set for workers from the 3rd to the 10th deciles, and in most years the majority of the minimum wages fall near the middle, in the 4th to 6th deciles of the distribution of wages.

Second, there are a significant number of workers who earn below the minimum wage. Table 3 shows this more clearly: Over the 1988–99 period, an average of 35.7% of all private sector workers earned less than 0.95 of the minimum wage. This proportion is higher for workers in small firms as compared to large firms, and in the rural areas compared to the urban ones. The lowest proportion is for the self-employed. Hence, minimum wages in Costa Rica clearly do not appear to act as a wage floor for a large number of workers, which signals incomplete enforcement. However, we know there is some effort on the part of the government to enforce minimum wages. Robles (2002) indicates that labor inspectors from the Ministry of Labor periodically check about 11% of the firms. The report by Robles notes that infractions of the minimum wage law were not the most important as they came in fourth among the 14 infractions listed. Finally, we know that if inspectors find employers guilty of not paying the minimum wages, they are fined and forced to pay the employee the back pay owed to him/her for up to two years. Nevertheless, we should also point out

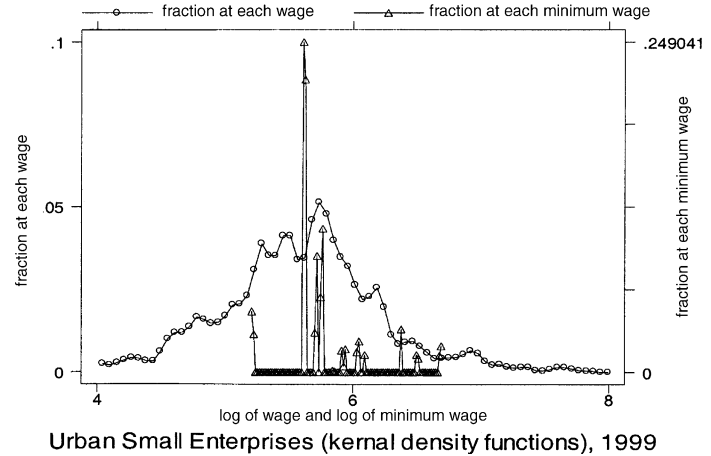
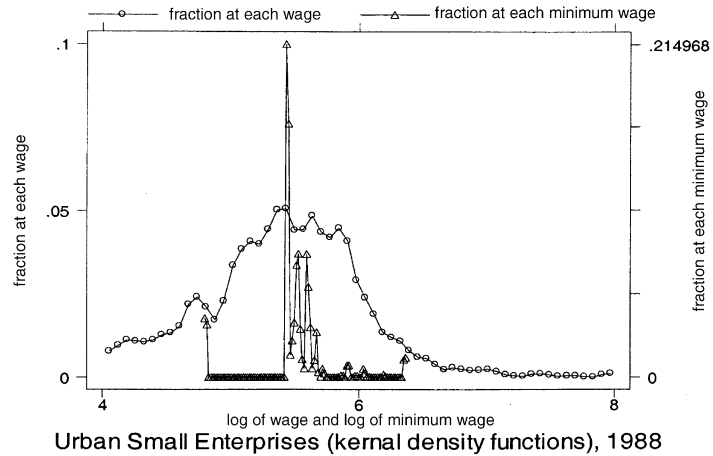
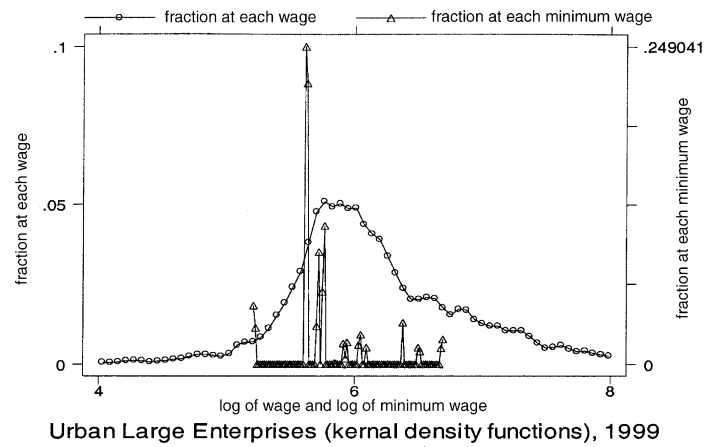
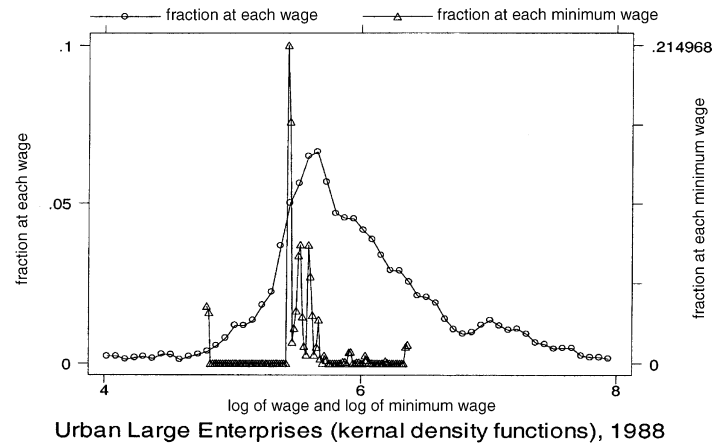


Figure 2. Comparing the distribution of minimum wages to the distribution of wages in the urban covered sector, 1988 and 1999.

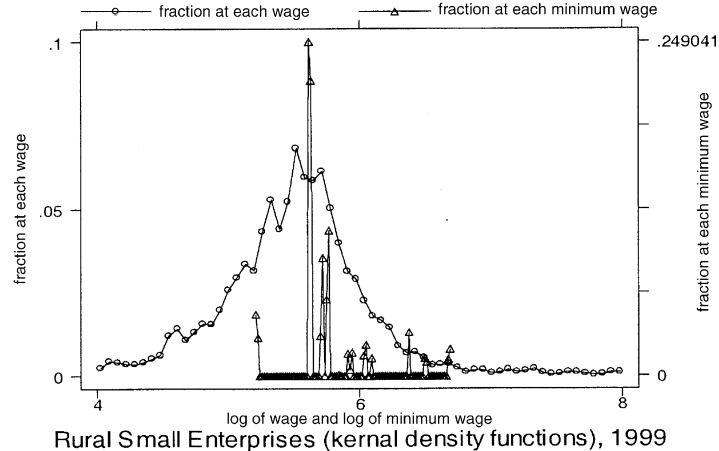
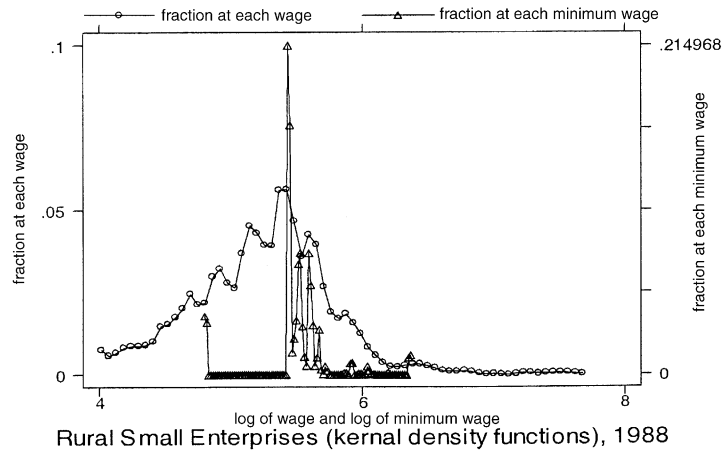
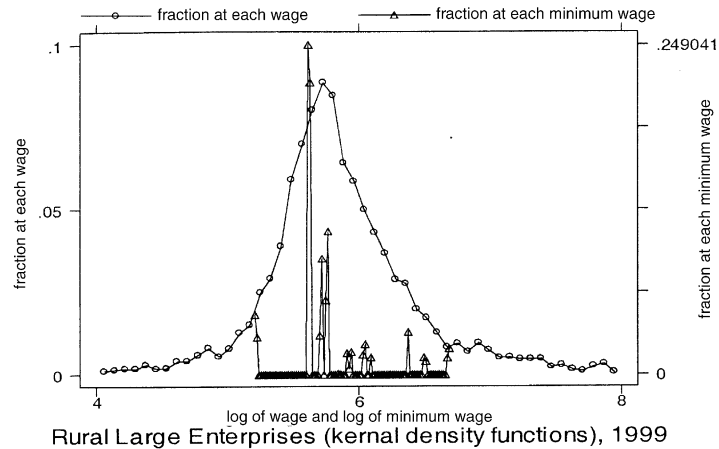
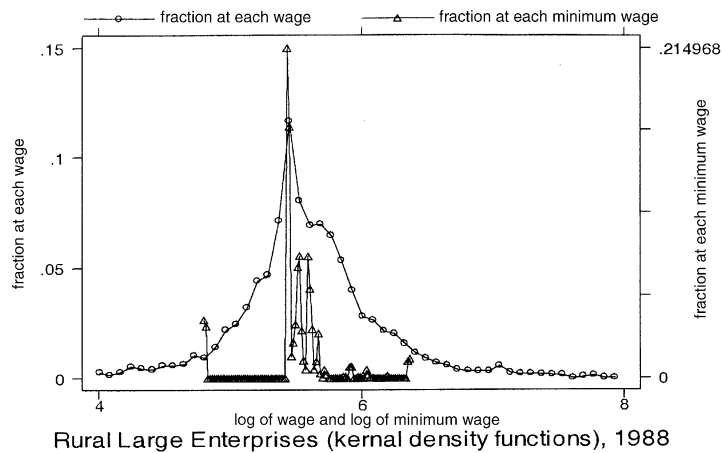


Figure 3. Comparing the distribution of minimum wages to the distribution of wages in the rural covered sector, 1988 and 1999.

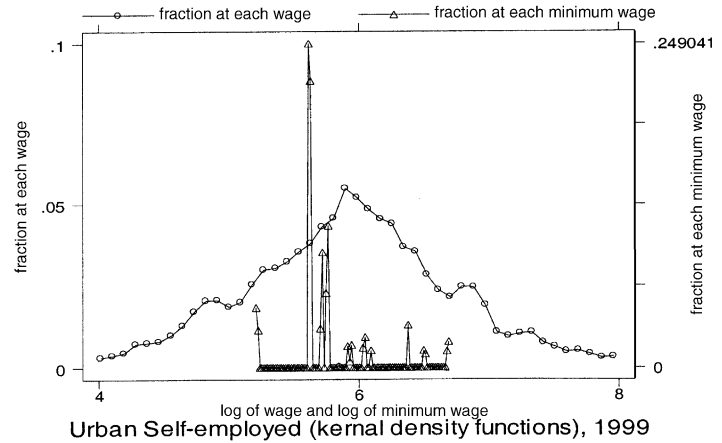
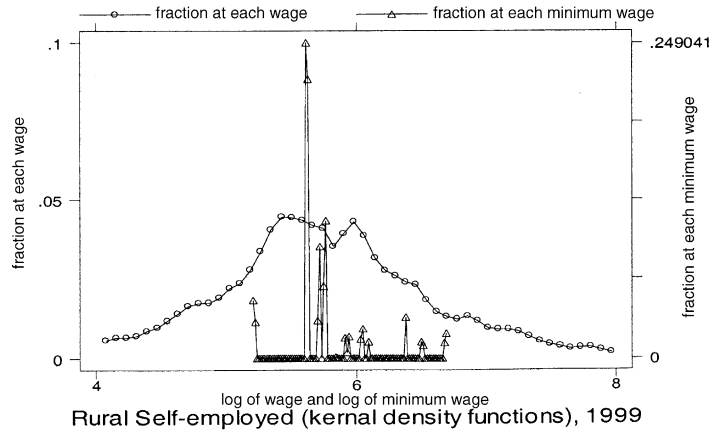
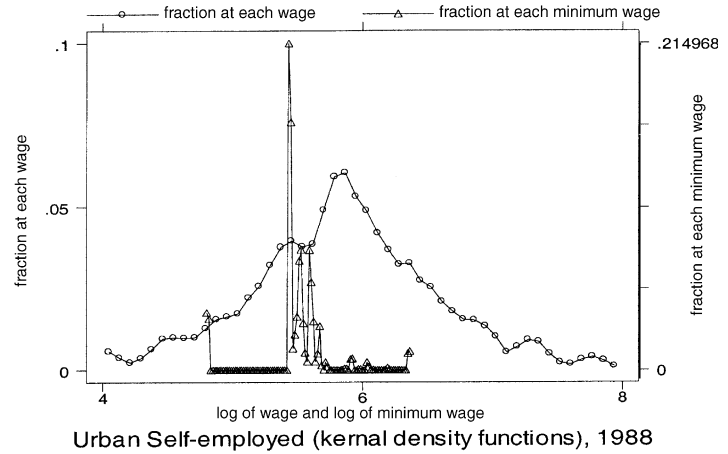
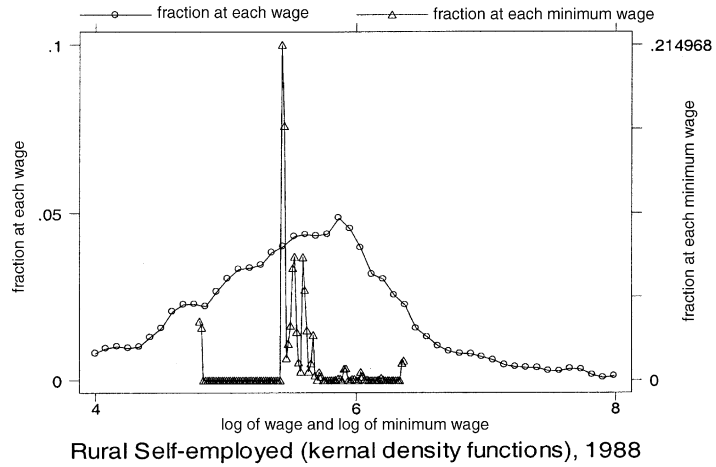


Figure 4. Comparing the distribution of minimum wages to the distribution of wages in the uncovered sector, 1988 and 1999.

Table 3. *Average proportion of workers earning at and below the minimum wage (1988–99)*

	Proportion earning 0.95–1.05 of minimum wage	Proportion earning below 0.95 of minimum wage
All sectors	0.088	0.357
Urban large firms	0.094	0.228
Urban small firms	0.082	0.407
Rural large firms	0.121	0.339
Rural small firms	0.100	0.539
Urban self-employed	0.051	0.285
Rural self-employed	0.053	0.425

Notes: Calculated from the *Costa Rican Household Survey*, using sample weighted data.

that the large proportion earning below the minimum wage is not simply a reflection of lack of enforcement. There may be some measurement error as well.

Nevertheless, [Figures 2 and 3](#) present evidence that minimum wages do affect the wages of some workers in Costa Rica. If legal minimum wages affect the actual wages of workers, then spikes in the distribution of minimum wages should be reflected with similarly located spikes in the distribution of wages. This appears to be the case in all sectors where workers are legally covered by minimum wages: the urban formal, rural formal, urban informal small firm, and rural informal small enterprise sectors. Indeed, the spikes are more marked among workers in small firms than in the large firm sectors. It is interesting to note that in both the rural and urban small firm sectors there is even a spike at the minimum wage for domestic servants.

Finally, [Figure 4](#) presents overlays of the distribution of minimum wages on the distribution

of actual hourly wages in 1988 and 1999 for the urban and rural uncovered sector. It provides evidence that minimum wages do not affect the wages of self-employed workers since there are no noticeable spikes in the distribution of self-employed earnings around the minimum wages. In sum, an examination of the graphs presented in [Figures 2–4](#) suggests that minimum wages do affect the wages of some paid employees in the four covered sectors, but do not affect the wages of self-employed workers (the uncovered sector).

(b) *Regression results*

The estimated coefficients on the minimum wage in Eqn. (1) are reported for each sector in [Table 4](#). The coefficients on the control variables are reported in [Table 5](#). The significant coefficients on minimum wages provide strong evidence that increases in the minimum wages cause increases in the wages of workers in the

Table 4. *Elasticity of wage with respect to minimum wage (dependent variable: log of hourly wage)*

Sector	Coefficient	Standard error	N	R ²
<i>Covered sectors</i>				
Urban large	0.108***	(0.030)	20,458	0.434
Urban small	0.152*	(0.093)	7,254	0.200
Rural large	0.164***	(0.038)	33,677	0.351
Rural small	0.398**	(0.112)	20,451	0.182
<i>Uncovered sectors</i>				
Urban self-employed	0.182	(0.144)	9,600	0.138
Rural self-employed	−0.052	(0.138)	21,843	0.155

Notes: See [Table 5](#) for the full set of regression coefficients. The data used in all regression are weighted using the sample weights. Estimated standard errors are robust to heteroskedasticity and are adjusted to account for potential clustering due to using a more aggregate variable (the minimum wage variables) with micro-level data.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

urban formal and both the rural large and rural informal small firm sectors. Among these sectors, the impact of minimum wages on average wages is lowest in the urban formal sector (with an elasticity of 0.108) and largest in the rural small enterprise sector (elasticity of 0.398). Intermediate elasticity estimates are found for the rural large enterprise sector (0.164) and the urban small firm sector (0.152). These findings are consistent with what we would expect if minimum wages were enforced similarly in all three sectors because there are more workers with wages near the minimum wage in the low-wage rural sectors. Hence, among covered sector workers, the finding of a relatively large elasticity for the rural small-scale sector implies that minimum wages are decreasing the wage differentials between this informal sector and the formal sector, *ceteris paribus*.

We find no evidence that minimum wages affect the wages of workers in the urban and rural self-employed (uncovered) sectors. The coefficients on the minimum wage variables in the two equations for self-employed workers are not significantly different from zero for any reasonable level of significance. These results are similar to those of Maloney and Núñez (2004), for the majority of the self-employed in Colombia. Unlike Fajnzylber (2001), however, we find no evidence of a "lighthouse effect." Our findings may not be surprising given our discussion of the ambiguous predictions from dual economy models of the impact of minimum wages on the uncovered sector. Is labor supply shifting out in the informal sector or is it shifting back in response to minimum wage increases? Since we do not have panel data, we cannot provide evidence on this question. However, in companion papers for Costa Rica (Gindling & Terrell, 2005a) and Honduras (Gindling & Terrell, 2005b) we show that the ranks of the unemployed grow as minimum wages are increased. This is consistent with the Mincer (1976) model and with the idea that the uncovered sector of employment is not necessarily the "residual sector."

5. FINAL REMARKS

Contrary to the assumption of much of the literature on labor markets in developing

countries, we show that legal minimum wages in Costa Rica have a significant effect on wages of employees in very small (five or fewer employees) as well as larger firms in both the urban and rural areas. This implies that the problem of differential enforcement is nowhere near as acute as suggested by the literature. We predict that this result is likely to be found in other Latin American countries as well.

Furthermore, the effect is larger for employees in small enterprises in both areas than for workers in large firms. The biggest positive impact of minimum wages is on rural small enterprises, the lowest wage workers. Hence, minimum wage enforcement contributes to reducing the wage gap between the rural informal sector and all other sectors that are covered by minimum wage legislation.

Legal minimum wages do not cover self-employed individuals, and we find no evidence that minimum wages affect the wages of the self-employed either with spikes in the distribution or with a significant coefficient in the wage regression. This is contrary to evidence from two studies on Brazil. We are confident that our results are not affected by the endogeneity problems that may have affected the earlier findings, but we also note that the theoretically predicted effect of minimum wages on the wages of workers in the uncovered sector is ambiguous and depends partially on whether the uncovered sector is a sector of last resort. We have evidence that leads us to conclude that the self-employed in Costa Rica are a heterogeneous group and not the residual sector for many unemployed workers. In fact, the higher average wage in this sector implies that it is a sector of choice for many. Nevertheless, our findings of positive effects of minimum wages on employees (covered sector) and no impact on the self-employed (uncovered) imply that an increase in the minimum wage raises wages of the employees relative to the self-employed.

Institutions and regulations play an important role in the labor markets of developing countries. This paper contributes to our understanding of the impact of institutions on the formal and the informal labor markets. Some of our findings challenge traditional development models and recent evidence from other countries; we expect this will provoke additional research in this area.

NOTES

1. Tauchen (1981) found some evidence in support of this theoretical prediction using data on the uncovered and covered sectors in nine regions in the United States.
2. This implies that a 1% increase in minimum wages raises the wages of paid employees by more than 1%, which strikes us as large and makes us question whether the minimum wage variable is not picking up other factors which affect both minimum wages and wages.
3. Public sector workers are excluded from the analysis since their wages are governed by a different set of decrees. Unfortunately, it is not possible to match individual observations in the Costa Rican household surveys across years to create panel data.
4. Other ways besides firm size of identifying the informal sector are possible however, as Ranis and Stewart (1999) write, "there is a strong correlation between smaller size enterprises and other criteria of informality" (p. 259).
5. See Maloney (2004) for evidence on the heterogeneity of work and income of the self-employed Latin America; House (1984) for evidence from Kenya; Earle and Sakova (2000) for Eastern Europe; Mead and Morrison (1996) and Yamada (1996) for evidence for several developing countries.
6. Kernel density estimates differ from the histograms in that the former allow the bandwidth to overlap and allow for different weighting schemes on the x 's. In particular, given a kernel $K(z)$, the estimated density function for x is

$$\hat{f}_k = \frac{1}{nh} \sum_{i=1}^n K\left[\frac{x - X_i}{h}\right],$$
 where n is the number of observations in the sample and h is the bandwidth. The points at which the density is estimated are indicated by x and the data by X_j . In all of our estimates, we evaluate the density at 100 points. In all of our estimates, the bandwidth chosen is 0.02 and the Epanechnikov kernel is used. In general, bandwidths are adjusted to be wider where there are few observations, as this allows for sharper fluctuations in the estimated density than seen in a normal by using less smoothing in ranges in which there are many observations.
7. These categories correspond, as best as we can make them, to the categories in the 1988 minimum wage legislation.
8. Minimum wages were set typically in January and July of each year, but sometimes they were set a little earlier or later.
9. Our description of the process of simplification of minimum wages in Costa Rica summarizes interviews with José Pablo Carvajal (Director, National Salaries Council) July 14, 2003; Yabera Alvarado (Planning Directorate, Ministry of Labor) July 15, 2003; and Pablo Sauma, July 9, 2003.
10. The t -statistics for the coefficient on the lagged minimum wage variable in the reduced form equation were 58 for the urban large firm sector, 14 for the urban small firm sector, 11 for the rural large firm sector, 34 for the rural small firm sector, 11 for the urban self-employed, and 12 for the rural self-employed.
11. The t -statistic for the coefficient on the residual was -0.47 for the urban large firm sector, -1.02 for the urban small firm sector, -1.06 for the rural large firm sector, 0.16 for the rural small firm sector, 5.12 for the urban self-employed, and -1.52 for the rural self-employed.

REFERENCES

- Bell, L. (1997). The impact of minimum wages in Mexico and Colombia. *Journal of Labor Economics*, 15(3), S102-S134.
- Castillo-Freeman, A., & Freeman, R. (1992). When the minimum wage really bites: The effect of the US-led minimum on Puerto Rico. In G. Borjas & R. Freeman (Eds.), *Immigration and the work force*. Chicago: University of Chicago Press.
- Earle, J. S., & Sakova, Z. (2000). Business start-ups or disguised unemployment? Evidence on the character of self-employment from transition economies. *Labour Economics*, 7(5), 575-601.
- Fajnzylber, P. (2001). Minimum wage effects throughout the wage distribution: Evidence from Brazil's formal and informal sectors. Department of Economics, Universidade Federal de Minas Gerais, Unpublished.
- Fields, G. S. (1975). rural-urban migration, urban unemployment and underemployment, and job-search activity in LDC's. *Journal of Development Economics*, 2(2), 165-187.

- Fiszbein, A. (1992). Do workers in the informal sector benefit from cuts in the minimum wage? World Bank Working Paper Series No. 826.
- Freeman, R. (1993). Labor market institutions and policies: Help or hindrance to economic development? In *Proceedings of the World Bank annual conference on development economics 1992* (pp. 117–144). Special Supplement to the World Bank Economic Review and World Bank Research Observer.
- Gindling, T. H., & Terrell, K. (2005a). Minimum wage effects throughout the labor market in Costa Rica. University of Michigan, Unpublished.
- Gindling, T. H., & Terrell, K. (2005b). Impacto de los Reajustes al Salario Mínimo en Honduras. Unidad de Apoyo Técnico, Secretaría de Estado en el Despacho Presidencial de Honduras.
- Harris, J., & Todaro, M. (1970). Migration, unemployment and development: A two sector analysis. *American Economic Review*, 60(1), 126–142.
- House, W. J. (1984). Nairobi informal sector: Dynamic entrepreneurs or surplus labor? *Economic Development and Cultural Change*, 32(2), 277–302.
- Lemos, S. (2002). The effects of the minimum wage on wages and employment in Brazil: A menu of minimum wage variables. University College London, Unpublished.
- Lewis, A. (1954). Development with unlimited supplies of labour. *The Manchester School*, 22(May), 139–192.
- Maloney, W. F. (2004). Informality revisited. *World Development*, 32(7), 1159–1178.
- Maloney, W., & Núñez, J. (2004). Measuring the impact of minimum wages evidence from Latin America. In J. J. Heckman & C. Pages (Eds.), *Law and employment: Lessons from Latin America and the Caribbean*. Chicago: The University of Chicago Press.
- McIntyre, F. (2004). An empirical model of informal markets, the minimum wage, and mandated non-wage benefits. Ph.D. Dissertation, Stanford University.
- McLeod, D., & Lustig, N. (1997). Minimum wages and poverty in developing countries: Some empirical evidence. In S. Edwards & N. Lustig (Eds.), *Labor markets in Latin America* (pp. 62–103). Washington: Brookings Institution.
- Mead, D. C., & Morrison, C. (1996). The informal sector elephant. *World Development*, 24(10), 1611–1619.
- Mincer, J. (1976). Unemployment effects of minimum wages. *Journal of Political Economy*, 84, 87–105.
- Ranis, F., & Stewart, F. (1999). V-goods and the role of the urban informal sector in development. *Economic Development and Cultural Change*, 47(2), 259–288.
- Ranis, G., & Fei, J. (1961). A theory of economic development. *The American Economic Review*, 51(4), 533–565.
- Robles, G. L. (2002). Denuncias por Infracciones Laborales Presentadas por la 'Inspección de Trabajo en Sede Judicial, Año 2001. Unidad de Investigación y Calidad de la Gestión, Dirección Nacional de Inspección de Trabajo, Costa Rica.
- Saint-Paul, G. (1994). Do labor market rigidities fulfill distributive objectives? Searching for the virtues of the European model. *IMF Staff Papers*, 41(4), 624–642.
- Strobl, E., & Walsh, F. (2003). Minimum wages and compliance: The case of Trinidad and Tobago. *Economic Development and Cultural Change*, 51(2), 427–450.
- Tauchen, G. E. (1981). Some evidence on cross-sector effects of the minimum wage. *Journal of Political Economy*, 89(31), 529–547.
- Todaro, M. (1969). A model of labor migration and urban unemployment in less developed countries. *American Economic Review*, 59(1), 138–148.
- World Bank (1990). *World Development Report*. New York: Oxford University Press.
- Yamada, G. (1996). Urban informal employment and self-employment in developing countries: Theory and evidence. *Economic Development and Cultural Change*, 44(2), 289–314.

APPENDIX A

See Table 5

(See overleaf)

Table 5. *OLS regressions, dependent variable: ln wage*

	Urban large scale	Urban small scale	Rural large scale	Rural small scale	Urban self-employed	Rural self-employed
Log of the minimum wage	0.108 (0.030)***	0.152 (0.093)*	0.164 (0.038)***	0.398 (0.112)***	0.182 (0.144)	-0.052 (0.138)
Schooling	0.045 (0.005)***	0.083 (0.010)***	0.039 (0.005)***	0.061 (0.010)***	0.010 (0.024)	0.045 (0.019)**
Experience	0.019 (0.009)**	0.071 (0.011)***	0.016 (0.006)**	0.050 (0.010)***	-0.017 (0.023)	0.020 (0.015)
Experience ² /1,000	-0.291 (0.355)	-1.496 (0.369)***	-0.027 (0.283)	-0.883 (0.348)**	0.492 (0.716)	-0.359 (0.436)
Experience ³ /10,000	0.026 (0.041)	0.091 (0.037)**	-0.032 (0.038)	0.037 (0.036)	-0.054 (0.066)	0.009 (0.039)
Gender	-0.063 (0.052)	0.044 (0.088)	0.025 (0.048)	0.194 (0.084)**	-0.483 (0.176)***	-0.193 (0.128)
Schooling · experience	0.001 (0.001)	-0.004 (0.001)**	0.001 (0.001)**	-0.001 (0.001)	0.002 (0.002)	0.000 (0.002)
Schooling · experience ² /1,000	-0.0595 (0.026)**	0.057 (0.051)	0.093 (0.023)***	-0.056 (0.041)	0.066 (0.066)	-0.035 (0.051)
Schooling · experience ³ /10,000	0.0007 (0.0003)**	-0.0002 (0.0006)	0.001 (0.0003)***	0.001 (0.0004)**	0.0006 (0.006)	0.0005 (0.0005)
Gender · experience	0.007 (0.006)	0.008 (0.009)	-0.538 (0.253)***	0.004 (0.008)	0.026 (0.017)	0.013 (0.011)
Gender · experience ² /1,000	0.006 (0.266)	-0.302 (0.335)	-0.538 (0.253)**	-0.153 (0.291)	-0.506 (0.573)	-0.167 (0.374)
Gender · experience ³ /10,000	-0.0002 (0.0003)	0.003 (0.003)	0.005 (0.003)	0.001 (0.003)	0.004 (0.005)	0.007 (0.036)
Schooling · gender	0.005 (0.003)	0.004 (0.007)	-0.003 (0.004)	-0.012 (0.005)**	0.017 (0.010)*	0.007 (0.008)
Constant	4.989 (0.198)***	3.745 (0.525)***	4.656 (0.334)***	1.537 (0.635)**	5.336 (0.832)***	5.649 (0.809)***
Dummies for minimum wage categories (350+)	Yes	Yes	Yes	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations	20,458	7,254	33,677	20,451	9,600	21,843
R ²	0.43	0.20	0.35	0.18	0.14	0.15

Notes: Estimated from the *Costa Rican Household Survey* sample weighted data.

* Significant at the 10% level.

** Significant at the 5% level.

*** Significant at the 1% level.

Available online at www.sciencedirect.com

